

Fraud Detection Analysis Using Customer Transaction Data

Project Overview

Financial fraud continues to pose significant risks to businesses, financial institutions, and customers worldwide. This project focuses on analyzing transactional payment data to identify patterns associated with fraudulent activities and provide insights that can improve fraud detection systems.

The dataset contains customer transaction records collected and structured by a Fraud Detection Team. It includes customer demographics, transaction details, merchant information, and fraud indicators. The primary objective of this analysis is to uncover trends, identify high-risk transaction characteristics, and support the development of data-driven fraud prevention strategies.

Business Problem

Fraudulent transactions can result in substantial financial losses, reputational damage, and reduced customer trust. Traditional fraud detection methods often struggle to identify sophisticated fraudulent activities in real time.

This project aims to answer the following questions:

- Which customer groups are most associated with fraudulent transactions?
- Are certain transaction categories more vulnerable to fraud?
- Does transaction amount influence fraud occurrence?
- Are there seasonal or monthly patterns in fraudulent activities?
- Which merchants experience the highest fraud rates?

The findings will help stakeholders strengthen fraud monitoring systems and improve risk management processes.

Dataset Description

The dataset consists of payment transactions made by various customers over a simulated four-month period.

Features

Step

Represents the month from the start of the simulation.

Step	Month
0	May
1	June
2	July
3	August

Customer

Unique identifier assigned to each customer.

Age

Categorized age groups:

Value	Age Range
0.0	≤ 18
1.0	19 – 25
2.0	26 – 35
3.0	36 – 45
4.0	46 – 55
5.0	56 – 65

Gender

Code	Gender
F	Female
M	Male

PostcodeOrigin

Origin postcode of the customer.

Merchant

Unique identifier representing the merchant involved in the transaction.

Category

Purchase category associated with the transaction.

Amount

Monetary value of the transaction.

Fraud

Target variable indicating transaction legitimacy.

Value	Meaning
0	Non-Fraudulent Transaction
1	Fraudulent Transaction

Project Objectives

The primary objectives of this project are:

1. Analyze customer transaction behavior.
2. Identify characteristics associated with fraudulent transactions.
3. Examine fraud distribution across demographic groups.
4. Evaluate transaction categories with elevated fraud risks.
5. Investigate seasonal trends in fraud occurrences.
6. Develop actionable recommendations for fraud prevention.

Methodology

Data Collection

The dataset was provided by the Fraud Team and contains structured transaction records collected during a four-month simulation period.

Data Preparation

The following preprocessing activities were conducted:

- Data quality assessment
- Missing value validation
- Data type verification
- Fraud label validation
- Category standardization
- Aggregation and summarization for reporting

Analytical Techniques

The analysis involved:

- Descriptive Statistics
- Fraud Rate Analysis
- Customer Segmentation
- Transaction Category Analysis
- Monthly Trend Analysis
- Merchant Risk Assessment

Tools Used

- Microsoft Power BI
 - Microsoft Excel
 - SQL (optional)
 - Python (optional)
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Key Performance Indicators (KPIs)

The following KPIs were developed to measure fraud performance:

Total Transactions

Total number of recorded transactions.

Total Fraudulent Transactions

Number of transactions classified as fraud.

Fraud Rate (%)

Fraud Rate = (Fraudulent Transactions ÷ Total Transactions) × 100

Total Transaction Amount

Combined value of all transactions.

Fraud Loss Amount

Total value of fraudulent transactions.

Average Transaction Amount

Mean transaction value across all transactions.

Analysis Framework

Customer Demographics Analysis

Purpose: Determine whether age or gender influences fraud likelihood.

Metrics:

- Fraud count by age group
- Fraud count by gender
- Fraud rate per demographic segment

Expected Outcome: Identification of customer groups requiring additional monitoring.

Transaction Amount Analysis

Purpose: Investigate whether fraud is concentrated within specific transaction ranges.

Metrics:

- Average fraud amount
- Maximum fraud amount
- Fraud distribution by amount

Expected Outcome: Detection of unusual transaction sizes associated with fraud.

Monthly Fraud Trend Analysis

Purpose: Monitor fraud behavior across the simulation period.

Metrics:

- Monthly fraud count
- Monthly fraud rate
- Monthly fraud losses

Expected Outcome: Identification of seasonal or monthly fraud patterns.

Merchant Fraud Analysis

Purpose: Determine which merchants are most frequently associated with fraudulent transactions.

Metrics:

- Fraud count per merchant
- Fraud loss by merchant
- Merchant fraud rate

Expected Outcome: Identification of high-risk merchants for enhanced monitoring.

Category Fraud Analysis

Purpose: Evaluate fraud occurrence across transaction categories.

Metrics:

- Fraud count by category
- Fraud value by category
- Category fraud percentage

Expected Outcome: Understanding of categories most vulnerable to fraudulent activity.

Expected Insights

The analysis is expected to reveal:

- Customer demographics most exposed to fraud.
- Categories generating the highest fraud incidents.
- High-risk merchants requiring investigation.
- Fraudulent transaction amount patterns.
- Monthly variations in fraud occurrence.
- Opportunities to strengthen fraud detection controls.

Recommendations

Based on analytical findings, organizations should consider:

Strengthening Monitoring Rules

Implement automated alerts for high-risk transaction categories and merchants.

Transaction Threshold Controls

Flag unusually large transactions for additional verification.

Customer Risk Profiling

Develop customer risk scores using age, transaction history, and behavioral patterns.

Merchant Risk Assessment

Regularly review merchants exhibiting elevated fraud rates.

Real-Time Fraud Detection Models

Leverage machine learning techniques to identify suspicious activities before transaction completion.

Continuous Fraud Auditing

Conduct periodic reviews of fraud trends and update detection strategies accordingly.

Conclusion

This project demonstrates the application of data analytics in identifying fraudulent transaction patterns and supporting fraud prevention initiatives. Through demographic analysis, transaction monitoring, category evaluation, and merchant risk assessment, organizations can gain valuable insights that improve fraud detection accuracy and reduce financial losses.

The findings generated from this analysis can serve as a foundation for advanced fraud detection systems, predictive analytics models, and real-time fraud monitoring frameworks.